JAGAT GURU NANAK DEV PUNJAB STATE OPEN UNIVERSITY, PATIALA

# DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS

M.Sc. (Computer Science) — Semester-IV

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# M.Sc. (Computer Science) Semester-IV Master Portfolio

## Jagat Guru Nanak Dev Punjab State Open University, Patiala (JGND PSOU)

**Academic Year: 2026 | Degree: Master of Science in Computer Science (M.Sc. CS)**

## 📌 University Course Inventory & Marks Matrix

| Course Code | Course Name | Credits | External Marks | Internal Marks | Total Mar |
|---|---|---|---|---|---|
| MSCS-4-03P | Major Project | 10 | 140 | 60 | 200 |
| MSCS-4-04 | Seminar | 2 | 0 | 50 | 50 |
| TOTAL | Final Semester Portfolio | 12 | 140 | 110 | 250 |

## 🏛️ Project & Research Alignment (SSR ONE AI)

Both courses are systematically anchored in the enterprise codebase `SSR One AI` (`e:/2026/ssr_one_ai`), an advanced multi-tenant hospitality SaaS POS and ERP system.

**Major Project (MSCS-4-03P):**

**Approved Title**: *"Design and Implementation of a High-Throughput, Offline-First, Multi-Tenant SaaS Enterprise POS Architecture with Integrated AI Copilot and Computer Vision OCR"*

**Scope**: Full-stack enterprise architecture encompassing React 19, Dexie.js IndexedDB, FastAPI ASGI, PostgreSQL 16 Row-Level Security (RLS), WebSockets, and Tesseract OCR.

**Seminar (MSCS-4-04)**:

**Approved Topic**: *"Architectural Patterns for Offline-First Distributed Web Applications: Balancing Zero-Latency User Experience with Eventual Consistency"*

**Scope**: Theoretical and practical research on local-first web storage, asynchronous reconciliation queues, idempotency tokens, and distributed conflict resolution.

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## 📂 Repository Directory Structure (`msccs/`)

```
e:/2026/ssr_one_ai/msccs/
├── 00-MASTER-INDEX.md                          # Master Portfolio & University
│
├── 01-MAJOR-PROJECT-MSCS-4-03P/                # Course Code: MSCS-4-03P (200 I
│   ├── 01-TOPIC-APPROVAL-APPLICATION.md       # Formal Topic Application to Co
│   ├── 02-PROJECT-SYNOPSIS.md                  # University-Mandated 8-Part Sy
│   ├── 03-FINAL-PROJECT-REPORT-THESIS.md       # University-Mandated 9-Part Pr
│   ├── 04-PRESENTATION-SLIDES.md               # 10-15 Min Viva Presentation D
│   ├── 05-VIVA-VOCE-DEFENSE-GUIDE.md           # Comprehensive External Examin
│   └── 06-RESEARCH-PAPER-IEEE.md               # Academic Research Paper Draft
│
└── 02-SEMINAR-MSCS-4-04/                       # Course Code: MSCS-4-04 (50 Ma
    ├── 01-SEMINAR-TOPIC-APPROVAL-APPLICATION.md # Formal Seminar Topic Applic
    ├── 02-SEMINAR-REPORT.md                    # 10-15 Page Academic Seminar R
    ├── 03-SEMINAR-PRESENTATION-SLIDES.md       # 15-20 Min Oral Presentation S
    └── 04-SEMINAR-VIVA-QA.md                   # Question Handling & Discussio
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## 📅 University Deadlines & Submission Milestones

| Milestone | University Window | Document to Submit | Target Deadline |
|---|---|---|---|
| **Topic Approval** | Beginning of 4th Semester | Topic Application Letters (Project + Seminar) | Week 1 |
| **Project Synopsis** | Within 15 Days of Approval | `02-PROJECT-SYNOPSIS.md` (8-Part Format) | Week 3 |
| **Seminar Report** | Within 30 Days of Approval | `02-SEMINAR-REPORT.md` (10–15 A4 Pages) | Week 5 |
| **Seminar Presentation** | Scheduled by Coordinator | `03-SEMINAR-PRESENTATION-SLIDES.md` (15–20 Min) | Week 6 |
| **Final Project Report** | At least 1 week before Viva | `03-FINAL-PROJECT-REPORT-THESIS.md` (30–35 Pages) | Week 10 |
| **Major Project Viva** | End of Semester | Presentation (10–15 Min) + Viva Discussion (7 Min) | Week 12 |
| **Paper Publication** | Semester Concluding Period | `06-RESEARCH-PAPER-IEEE.md` submitted to Conference / Journal | Post-Viva |